

SoftWall: Certifying Conditional Recovery for Shared-GPU AI-RAN

ANONYMOUS AUTHOR(S)

AI-and-RAN platforms seek to share a GPU between radio processing and external AI. NVIDIA Multi-Process Service (MPS) enables concurrent execution, but it does not define a temporal contract for the radio path. Optional neural reception introduces a further dependency: until the neural outcome is known, the same transport block retains a conventional-recovery obligation that is absent from the current GPU queue.

We present SoftWall, a runtime substrate that represents each unresolved obligation as recovery debt and maintains an executable all-fail certificate. SoftWall refines this certificate through endpoint admission, atomic outcome application, external-AI leasing, distributed ownership, and physical CUDA completion. It also predicts a provenance-qualified feasibility envelope with four outcomes: safe and useful, safe without AI slack, mandatory infeasible, and unqualified.

Two prespecified boundary states show that current idleness cannot provide this contract: debt-blind admission can exceed the declared radio guard by 1-12 ms, while SoftWall rejected all 60 corresponding requests before GPU launch. On two independent four-A100 nodes, a qualified warm path executed 4,800 actual TensorRT NeuralRx requests, 1,312 shared-cuPHY recoveries, 1,077 Qwen units, and 4,800 radio commits without an observed deadline or invariant violation. The model matched an exact checker on 16,023 qualified states and predicted all 180 prespecified physical boundary rounds; a 64-debt certified scheduler reached 1.494 ms p99 decision latency. A certificate-preserving recovery-first baseline matched SoftWall at 385,262 timely tokens, and the full-bound sensitivity improved by only 0.066%, below the prespecified 5% effect threshold. We therefore make no optimizer claim. These results qualify finite-sample synthetic modes rather than production HARQ: the best 4.5 ms testMAC-target prototype passed only 995/1,000 development requests, and External Aerial TDL-A remains unqualified. A separate Sionna CDL-D/E holdout establishes conditional NeuralRx value only for its declared channel mode.

CCS Concepts: • **Computer systems organization** → **Distributed architectures**; **Real-time systems**; • **Networks** → *Network resources allocation*.

Additional Key Words and Phrases: AI-RAN, GPU scheduling, neural receiver, conditional recovery, MPS, feasibility certificate

ACM Reference Format:

Anonymous Author(s). 2027. SoftWall: Certifying Conditional Recovery for Shared-GPU AI-RAN. In *Proceedings of ACM SIGMETRICS 2027 (SIGMETRICS '27)*. ACM, New York, NY, USA, 20 pages. <https://doi.org/10.1145/nnnnnnnn.nnnnnnn>

1 Introduction

AI-and-RAN consolidates radio physical-layer processing and external inference on shared accelerators. The consolidation can reclaim idle GPU capacity, but it also couples a deadline-sensitive radio path to kernels whose arrivals and execution times follow a different service objective. Existing systems allocate GPU fractions, place requests across accelerator pools, reclaim reserved capacity, and split background work into bounded units [1, 10, 11, 13]. The remaining problem is not another allocation knob. It is a future mandatory job created by optional reception.

Consider one uplink transport block (TB). A timely NeuralRx result may improve its decoding outcome, while failure, lateness, or lost evidence leaves conventional decoding mandatory. Before that outcome is known, the conventional path is absent from the execution queue but remains part of the radio contract. Several cells can hold this obligation at once, and several RAN homes can send recovery to the same cuPHY lane. A scheduler that sees an idle GPU can therefore admit external AI into time that must remain available for correlated recovery.

The obvious conservative response is to reserve every recovery until every neural outcome resolves. That response preserves safety but discards capacity released by successful outcomes. The useful middle ground requires a runtime to answer three questions together: which speculative paths may run, which conventional jobs must remain executable under all-fail behavior, and which external-AI unit can finish without invalidating that recovery plan. Queue occupancy, MPS percentage, and local per-home admission do not represent all three.

SoftWall answers these questions with *recovery debt*. Every unresolved optional path owns one debt, and the runtime maintains a global executable all-fail certificate over those debts. A success deletes one debt; failure, lateness, or missing evidence retains it. External AI receives a lease only when a candidate recovery calendar and the complete AI transaction pass an independent verifier. This construction makes MPS an execution mechanism, not the contract.

A mathematical certificate alone is insufficient. In our implementation, a valid calendar became stale before kernel launch, simultaneous outcomes produced order-dependent intermediate states, a fixed lease omitted launch control, and reply loss left ownership ambiguous. SoftWall carries the same certificate through CUDA, IPC/P2P, and distributed-control state using bounded revalidation, atomic outcome batches, physical latest-start enforcement, generation-safe ownership, and fence-confirmed retirement. These mechanisms are established individually. Their combination refines an AI-RAN recovery invariant from admission to physical completion.

The paper makes three connected contributions, followed by a physical validation:

- (1) **Certificate-to-physical-execution refinement.** We derive seven requirements from measured counterexamples and bind a global recovery certificate to endpoint, transport, AI lease, worker epoch, completion fence, and single-commit state.
- (2) **Predictive feasibility envelope.** We classify each qualified state as qualified-safe-useful (QSU), qualified-safe-no-slack (QSN), mandatory-infeasible (MI), or unqualified (UQ) before physical execution. A candidate scheduler never controls admission until an independent verifier accepts its certificate.
- (3) **AI-RAN recovery-debt model.** We formalize same-TB optional reception, prove that one all-fail certificate covers every allowed failure subset, and show why current idleness cannot establish the radio contract.
- (4) **Evaluation with negative results.** We exercise actual TensorRT NeuralRx, Aerial/cuPHY recovery, Qwen inference, CUDA IPC, NVLink P2P, MPS, and distributed faults. Strong safe baselines show no material AI-throughput headroom in the evaluated trace, which separates the substrate claim from an optimizer claim.

Claim boundary. The guarantee is conditional on the qualified bounds and provenance of the evaluated mode. SoftWall does not claim that MPS provides hard isolation, that finite-sample maxima are worst-case execution times (WCETs), or that the synthetic timing contract is a production HARQ deadline. Five lifecycle subsets are qualified or partially qualified and five remain UQ; the claim does not require or imply a 10/10 lifecycle matrix.

Artifact plan. Upon acceptance, we will release the controller and verifier, finite-state and exact model checkers, machine-readable protocols and analyzers, sanitized timing records, and manifests that bind source to reported results through a long-lived archival repository. Proprietary Aerial/cuPHY, TensorRT, and model components will be accompanied by acquisition and build instructions rather than redistributed.

2 Background and Problem

2.1 AI-and-RAN on a shared GPU

A shared accelerator exposes radio and external-AI kernels to the same execution and lifecycle state. The hardware mechanism determines where kernels may run; a radio contract must additionally determine which future work must remain executable.

MPS Execution. MPS permits CUDA processes to execute concurrently and can limit a client's active-thread percentage [7]. It does not bound host scheduling, queueing, context lifecycle, allocation, or arbitrary kernel blocking. Our diagnostics observe deadline misses under both tested caps and separate the effect from a CPU-only sham in Section 8.1. We therefore use MPS to create concurrency and place the admission contract above it.

MIG-Off Reclamation. MIG provides spatial separation, but fixed partitions prevent one tenant from reclaiming another partition's conditional capacity. We intentionally study MIG-off modes in which radio and external AI can share qualified slack. A missing or failed bound classifies the mode as UQ; the runtime never imports a bound from another placement or lifecycle.

2.2 Optional NeuralRx creates future mandatory work

An optional receiver is speculative radio execution. It can improve the TB outcome, but it cannot remove the requirement to produce a conventional result after failure, lateness, or lost evidence.

Same-TB Dependency. Before the NeuralRx outcome resolves, conventional decoding is both conditional and mandatory under an allowed future branch. It need not appear in a GPU queue. Static worst-case reservation protects this branch but retains the reservation after timely neural success until the controller observes and commits that outcome.

Shared Recovery. Several cells can select NeuralRx together, and their failures can be correlated. Several RAN homes can also send conventional work to one shared cuPHY lane. Per-home schedules do not compose on that lane; the runtime must schedule the union of unresolved obligations.

External-AI Admission. Qwen is a representative external workload that creates physical GPU contention. It is not an AI-RAN-domain workload. Its request may run before unresolved recovery only when the complete transaction, including launch control and physical completion, leaves an executable recovery plan.

Figure 1 shows the shared recovery path and one evaluated decision-boundary example. It separates optional reception, conventional recovery, external AI, and the certificate that links their physical completion.

2.3 Threat model and qualification

SoftWall provides a conditional guarantee for an explicitly qualified mode. Qualification binds service bounds to the environment in which they were measured and defines the events the runtime must contain.

Mode Provenance. A mode fixes GPU topology, MPS settings, process lifecycle, worker epoch, memory residency, co-run class, software, and finite-sample whole-path bounds. Changing any element requires requalification. The synthetic release period and result expiry used in evaluation are harness parameters, not NR slot or production HARQ timing.

Contained Events. The runtime handles NeuralRx failure and lateness, stale and duplicate messages, bounded reply loss before or after a control-plane effect, missing AI replies after physical completion, and same-worker channel reconnection. It fails closed when terminal physical evidence is absent.

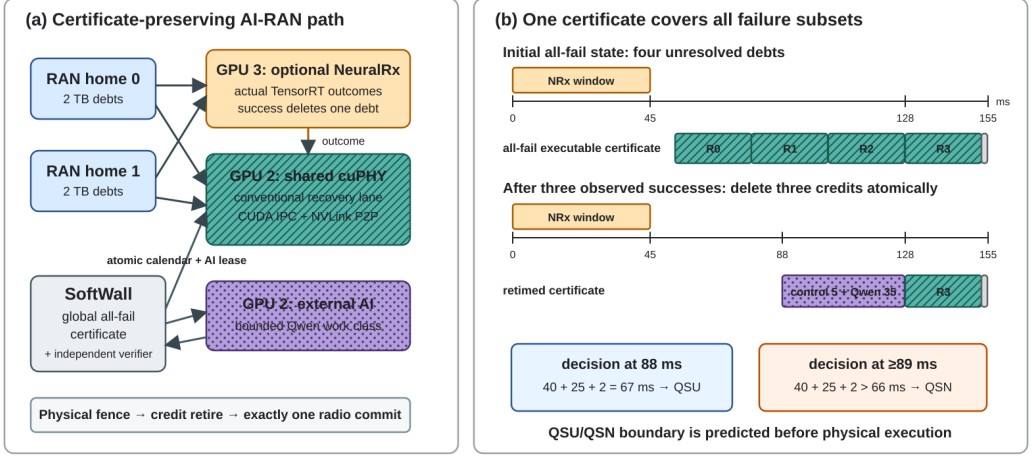


Fig. 1. Certificate-preserving AI-RAN path and one qualified boundary example. The all-fail schedule covers unresolved recovery, observed successes delete debt atomically, and external AI is admitted only when the remaining certificate stays executable.

Excluded Events. Arbitrary GPU hangs, driver reset, unbounded host suspension, production-clock uncertainty, and unqualified workloads remain outside the claim. Mandatory offered load without an all-fail schedule is MI rather than a reason to weaken the safety check.

3 Failure-Derived Design Requirements

SoftWall uses known primitives: bounded admission, list scheduling, generation checks, CUDA fences, IPC/P2P rings, and at-most-once ownership. The contribution is the refinement chain that connects these primitives to one conditional-recovery invariant. Table 1 names the counterexamples and defers all measurements to Section 8.

From Local to Global. Recovery feasibility belongs to the contended physical lane. The certificate must cover the union of obligations before optional endpoint or external-AI admission.

From Decision to Launch. A certificate must remain current through control delay, outcome observation, and CUDA launch. A lease therefore includes launch control and carries an absolute latest-start enforced by the worker.

From Logical Completion to Physical Retirement. RPC success is not CUDA completion, and reply loss is not proof that no effect occurred. Generations, terminal markers, and physical fences bind ownership to retirement. The model in Section 4 defines this invariant, and the design maps it to executable components.

4 Model

The model separates what the controller knows from what physical execution may still require. It represents conditional work, qualified service, shared resources, and terminal evidence in one state.

4.1 Jobs, modes, and provenance

A job exposes only information available at admission. Runtime decisions cannot use future CRC, future arrivals, realized service time, or a hidden channel label.

Table 1. Counterexamples define the certificate-to-execution refinement chain.

Counterexample	Broken assumption	Required refinement
Local-safe/global-unsafe	Local certificates compose on a shared lane	Global obligation set and certificate
Dispatch tail	A certificate stays fresh until launch	Bounded revalidation and physical latest-start
Sequential cutoff replay	Simultaneous outcomes may apply in any order	Atomic outcome batch
Fixed AI lease	Execution time covers launch control	Control charge and AI blackout
Reply loss after effect	Retrying ambiguity is harmless	Generation quarantine and fail-closed admission
Two outstanding pre-pares	RPC at-most-once implies visible ownership	Single unlaunched-token invariant
Warm-bound reuse	Lifecycle does not affect service tails	Provenance-scoped qualification

Radio Jobs. A radio request is

$$J_i = (r_i, d_i, h_i, c_i, K_i, q_i), \quad (1)$$

where r_i is release, d_i result expiry, h_i recovery home, c_i the qualified conventional host-to-commit bound, K_i eligible NeuralRx endpoints, and q_i optional radio value. Endpoint eligibility includes its queue, transport, observation, and commit path under the active mode.

External-AI Jobs. An external-AI request is

$$A_a = (u_a, v_a, \ell_a, b_a, g_a), \quad (2)$$

where u_a is visible arrival, v_a timely value, ℓ_a deadline, b_a the qualified physical-completion bound of its work class, and g_a the guard before recovery. The controller may select only from requests visible at the current event.

Qualified Modes. Every bound carries provenance

$$\pi = (\text{GPU}, \text{placement}, \text{MPS}, \text{lifecycle}, \text{co-run}, \text{software}). \quad (3)$$

A bound is admissible only when π matches the active runtime. This rule prevents a warm or local measurement from qualifying a cold, restarted, or remote path.

4.2 Recovery debt and all-fail certificates

An unresolved optional path creates work that may become mandatory. Recovery debt records that work before any conventional kernel enters the queue.

Debt. While outcome i is unresolved, it induces

$$R_i = (a_i(t), d_i - g_i, c_i, h_i), \quad (4)$$

where $a_i(t)$ is the earliest legal recovery start at state time t . For unresolved set $U(t)$, an all-fail certificate assigns every R_i an interval $[s_i, s_i + c_i]$ that satisfies release, expiry, resource capacity, and non-overlap. Any AI blackout or transport use that contends with recovery is part of the same schedule.

All-Fail Dominance.

PROPOSITION 1 (ALL-FAIL DOMINANCE). *If a successful NeuralRx only deletes its conventional recovery and does not increase any remaining job's demand, a feasible certificate for $U(t)$ is feasible for every failure subset $F \subseteq U(t)$.*

PROOF. Delete the intervals of successful requests from the all-fail schedule. Releases, expiries, capacities, and non-overlap of every retained interval remain unchanged. $\square$

Applicability. The reduction avoids enumerating $2^{|U|}$ outcome subsets. It does not apply if success creates new mandatory work or changes another job's service bound. Such a mode requires a new certificate and qualification.

4.3 Global composition and atomic state

The resource on which recoveries execute determines certificate scope. Logical ownership and physical retirement must change with that certificate as one state transition.

Composition. Certificates over disjoint recovery lanes compose by union. Certificates from homes that share one lane do not compose independently. SoftWall therefore builds a global certificate over the union of their obligations.

State Transaction. The runtime state is

$$X_t = (S_t, Q_t, T_t, L_t, E_t, C_t, G_t), \quad (5)$$

where S is the recovery schedule, Q endpoint credits, T transport generations, L AI leases, E fences and worker epochs, C radio commit state, and G multi-home AI ownership. SoftWall constructs a private candidate X' and commits it only when

$$\begin{aligned} \text{VALID}(X') \equiv & \text{ExecutableRecovery}(S') \wedge \text{UniqueCredits}(Q', T') \\ & \wedge \text{TimelyAI}(L', E') \wedge \text{CurrentGenerations}(X'). \end{aligned} \quad (6)$$

A failed check leaves X_t unchanged. A lease retires only through matching physical completion or a qualified non-launch record.

Safety Invariant.

SAFETY INVARIANT 1 (CERTIFICATE-TO-EXECUTION REFINEMENT). *For every admitted radio request, every allowed unresolved-outcome branch retains a collision-free conventional path completing by expiry. No endpoint, transport, AI, or output credit is reused before matching terminal physical evidence, and each request commits at most one radio result.*

4.4 Decision window and necessity

The certificate defines the time window in which external AI may execute. The window depends on future mandatory work, even when the physical lane is idle now.

Safe AI-First Placement. Suppose m_h equal-length recoveries are compacted at expiry d_h with guard g_h . Let b_{eff} include synchronous control, AI service, and completion guard. A sufficient condition for AI-first execution is

$$T_{\text{dec},h} + b_{\text{eff}} \leq d_h - g_h - m_h c_h. \quad (7)$$

The right side is the first recovery start retained by the certificate. This condition is mode-specific and sufficient; it is not a general online optimum.

Current Idleness.

PROPOSITION 2 (CURRENT IDLENESS IS INSUFFICIENT). *Consider one non-preemptive recovery lane, common guard boundary D , unresolved recovery bounds $c_1, \dots, c_m$, and AI transaction bound b_{eff} . If*

mandatory-only execution is feasible but

$$t + b_{\text{eff}} + \sum_{i=1}^m c_i > D, \quad (8)$$

then a policy that admits AI from current idleness while ignoring unresolved recovery cannot guarantee the radio contract.

PROOF. Choose the allowed realization in which the AI transaction and every recovery take their declared bounds. Non-preemptive execution places the last recovery after D by Equation 8. Physical idleness at t does not remove future mandatory demand. $\square$

Controller Choices. A conservative controller can preserve the initial certificate and drain recovery before AI. SoftWall permits AI-first execution only with a new witness satisfying Equation 7. The evaluation separates the necessity of this certificate from whether AI-first placement improves throughput.

5 Design

The design maps each term in the safety invariant to a runtime component. No component can admit, launch, retire, or commit work by consulting only its local queue.

5.1 Overall architecture

Figure 1 shows the architecture. The data path executes radio and external-AI work; the control path maintains the certificate and transfers authority between physical workers.

Radio Components. Each *RAN Home* owns request release, expiry, and single-commit state. *Optional NeuralRx* executes speculative TensorRT reception on an eligible endpoint. *Shared cuPHY* owns the conventional-recovery lane used by all homes. The shared lane is why certificate scope follows the physical resource rather than the home boundary.

Certificate Components. *SoftWall* stores recovery debt, calendars, endpoint and transport generations, leases, worker epochs, and commit state. An *Independent Verifier* accepts a candidate only after checking job identity, release, service, expiry, lane capacity, non-overlap, AI blackout, and current generation. The candidate scheduler cannot bypass this verifier.

External-AI Component. *External AI* receives a bounded work class, absolute latest-start, and lease generation. It returns a physical fence or a qualified non-launch marker. This terminal evidence permits credit retirement while keeping AI ownership and radio recovery in the same transaction.

5.2 Radio admission and outcome transition

The controller establishes a valid all-fail state before speculative execution and preserves it when outcomes arrive. This ordering prevents optional work from consuming capacity needed by the mandatory branch.

Mandatory-First Admission. On radio release, *SoftWall* reserves conventional recovery before considering *Optional NeuralRx*. An endpoint is eligible only when its whole-path bound fits the fallback cutoff and it owns a generation-matching ring and transport credit. Rejection leaves the conventional path intact.

Atomic Outcome Batch. All outcomes observed at a common cutoff enter one transition. Timely successes delete their debts; failure, lateness, and missing evidence retain debt. Sequential replay is forbidden because it can construct an intermediate state that never existed and evaluate it at a later host time.

Table 2. Predictive feasibility classes.

Class	Meaning
QSU	All-fail recovery is executable and at least one qualified AI unit is admissible.
QSN	All-fail recovery is executable, but the candidate AI unit cannot safely fit.
MI	The mandatory all-fail schedule is infeasible under known bounds.
UQ	A required bound or lifecycle provenance is absent or failed qualification.

Replanning. After applying the batch, *SoftWall* rebuilds the unresolved calendar from the current time. It may retime recovery intervals, but it cannot delete unresolved work or reuse a physical credit without terminal evidence.

5.3 AI lease and physical enforcement

AI admission is a certificate update rather than a queue insertion. The update covers control delay, physical execution, and the interval in which recovery must remain unobstructed.

Atomic Replan and Lease. *SoftWall* evaluates one candidate containing the retimed calendar and one AI lease. The candidate commits only when recovery remains executable, the AI transaction meets its deadline and recovery guard, credits are unique, and every generation is current. Failure restores the prior state without a partial lease or calendar update.

Launch-Time Revalidation. A certificate older than the qualified launch-control interval is discarded. *External AI* receives an absolute latest-start and emits a non-launch marker after that time. This rule closes the interval between host admission and CUDA launch.

Fence-Based Retirement. The executor performs only the next non-preemptive action named by the certificate. Matching CUDA completion retires buffers, transport credits, and leases. A host timeout without physical evidence quarantines ownership and blocks new optional admission.

5.4 Distributed ownership and fault containment

One global AI queue may serve several RAN homes. *SoftWall* keeps ambiguity local while allowing qualified mandatory radio work and unaffected homes to continue.

Single Unlaunched Token. Prepare creates at most one unlaunched token per home. Commit grants launch authority only after local and global certificates are current. A second prepare cannot create another broker-held token for the same ownership epoch.

Ambiguous Effects. Reply loss after a broker-side effect does not trigger a fresh retry. The home quarantines the token, closes new AI admission, and waits for a current terminal marker. Stale and duplicate replies cannot retire a newer generation.

Lifecycle Boundary. Prepare, abort, commit, and complete carry generations and bounded markers in the qualified same-worker reconnect mode. Full worker replacement and durable reconciliation require another mechanism and remain UQ.

5.5 Feasibility classification

Classification makes the contract operational before physical execution. Safety, usefulness, overload, and missing evidence are distinct outcomes.

Safe and Useful versus Safe without Slack. QSU and QSN share a valid radio certificate. They differ only in whether the visible AI candidate fits. This distinction exposes conditional capacity without weakening radio admission.

Infeasible versus Unqualified. MI is a proved scheduling result under known bounds. UQ reports insufficient authority to apply those bounds. SoftWall fails closed for UQ even when unqualified sample timing appears to fit.

6 Implementation

The prototype binds the model to persistent CUDA workers and explicit control objects. Its purpose is to make certificate state observable and enforceable across process and GPU boundaries.

6.1 Runtime topology

The implementation runs with MIG disabled and MPS enabled. RAN homes, neural endpoints, conventional recovery, and external AI execute in separate persistent workers.

Radio and AI Workers. Local and remote TensorRT NeuralRx workers hold preallocated bindings and CUDA graphs. Persistent Aerial/cuPHY workers execute conventional recovery, while a Qwen2.5-1.5B worker executes bounded prefill classes. Multi-home modes direct conventional work to one shared recovery GPU.

Transport. CUDA IPC rings connect processes on one GPU, and NVLink P2P rings connect GPUs. Each slot carries a request generation and physical completion state. Copy completion does not imply receiver or recovery completion; the owning worker publishes a separate terminal fence.

Controller. The controller stores recovery calendars, endpoint and transport credits, worker epochs, global AI ownership, and single-commit state. Unix-domain RPC carries control messages, while CUDA-resident data stays on IPC/P2P paths.

6.2 Certificate engine

The certificate engine separates schedule construction from admission authority. This separation contains candidate-scheduler bugs and makes each decision independently checkable.

Candidate Scheduler. The scheduler applies deterministic list orders to radio jobs and visible AI candidates. It emits a complete schedule, ownership delta, and mode fingerprint. Exact search is used only as a small-state oracle and is not an optimization contribution.

Independent Verifier. The verifier checks job identity, release, qualified service, expiry, lane capacity, overlap, AI blackout, credit uniqueness, and generation. The runtime cannot execute a candidate rejected by the verifier. This interface also supports retrospective comparison against exact feasibility.

Event Loop. The executor performs the first non-preemptive action in an accepted certificate, observes its terminal event, and replans. It never reorders physical work from an unverified host-side queue.

6.3 Measurement and mode separation

Instrumentation distinguishes logical decisions, host progress, GPU execution, and physical retirement. This distinction prevents a host timestamp from being reported as a kernel boundary.

Recorded Events. The runtime records release, admission, dispatch, worker observation, CUDA completion, commit return, request and payload identity, ring generation, worker epoch, and lease retirement. Nsight traces check GPU ordering for selected modes.

Python and Native Paths. The qualified C158-C162 path uses pyAerial with persistent workers. A later native C++ effort verifies input-layout and decoder parity only. It is not timing-qualified and its measurements are never mixed with the Python mode's service bounds.

Artifact Binding. Source hashes, protocols, seeds, node exclusions, analyzer versions, failures, and result hashes are preserved in manifests. Every empirical block in this paper maps to a machine-readable artifact and field set.

Table 3. Evaluation questions and their decision criteria.

RQ	Question	Decision criterion
1	Is MPS itself the contract?	Misses under cap-only diagnostics reject the premise.
2	Does the full path preserve the certificate?	Radio, recovery, AI, credit, and commit gates all pass.
3	Does refinement contain specified faults?	State audit and physical fault classes preserve invariants.
4	Does the envelope predict physical admission?	Exact, retrospective, and unopened-boundary decisions agree.
5	Does certified construction scale?	No false-safe result; decision and verification remain bounded.
6	Does AI-first retiming add timely value?	Prespecified effect threshold against a strong safe baseline.
7	Where does qualification end?	Lifecycle, production timing, and channel gates retain failures.

7 Evaluation Methodology

The evaluation tests the invariant before performance. It first asks whether the physical system obeys the qualified contract, then whether the model predicts admission, and finally whether the additional flexibility has measurable utility.

7.1 Questions and metrics

We organize the experiments around seven questions. Table 3 connects each question to the claim it can support.

Safety Metrics. We count radio expiry misses, declared-bound violations, guard crossings, invalid certificates, duplicate commits, stale generation acceptance, transport-credit reuse, and residual leases. Zero observed events is reported as a finite-sample result rather than a probability or WCET claim.

Utility Metrics. External-AI utility is deadline-compliant token value and completed request count. Radio utility is correct TB outcome under the declared channel mode. Candidate-policy comparisons preserve the same radio decisions and observable inputs.

7.2 Testbed, workloads, and contract

Authoritative campaigns use two independent nodes, each with four A100-SXM4-40GB GPUs connected by NVLink. Facility identity is withheld for anonymous review. MIG is off and MPS is on.

Radio and External AI. Radio execution uses NVIDIA Aerial/cuPHY and TensorRT NeuralRx. Qwen2.5-1.5B prefill creates external GPU contention. BurstGPT supplies arrivals and token values from four non-overlapping public-trace windows [15]; Qwen is not labeled an AI-RAN-domain workload.

Qualified Synthetic Contract. The evaluated warm mode uses harness period/expiry P180/D155, with finite-sample whole-path bounds $B_{\text{NRx}} = 45$ ms, $B_{\text{conv}} = 25$ ms, $B_{\text{AI}}(\text{context64}) = 35$ ms, and a 2 ms guard. The period is not an NR slot, and the expiry is not a production HARQ deadline. Other AI contexts use their separately qualified work class.

Channel Modes. Synthetic channels drive system qualification. A separate Sionna CDL-D/E mode tests radio value under its own channel, radio, model, and precision contract. Aerial TDL-A and field IQ remain outside that supported mode.

7.3 Protocols and baselines

Each authoritative campaign freezes its decision rule before opening the outcome. Development and holdout runs use independent seeds and nodes where the gate requires them.

Experimental Discipline. Protocols fix source hash, workload, bounds, seeds, node assignment, exclusions, analyzer, and stop rule. Failed, negative, and UQ runs stay in the ledger. A model-predicted boundary is frozen before physical execution.

Safe Baselines. *Static global-safe* retains worst-case recovery reservation. *Certificate-preserving recovery-first* uses the same max-radio decisions, initial all-fail admission, arrivals, bounds, workers, and single-commit rule as SoftWall, but drains unresolved recovery before external AI. SoftWall adds AI-first placement only with a valid witness. The offline oracle sees the realized trace and supplies an unattainable upper bound.

Diagnostic Baselines. Debt-blind current-idle admission, local-only certificates, and uncontrolled MPS diagnose broken assumptions. They are not safe performance comparators and are not used to claim throughput gains.

Reproducibility. The anonymous artifact package contains source, protocols, manifests, analyzers, exact checker, independent verifier, result summaries, and figure builders. Hardware campaigns require proprietary Aerial/cuPHY and TensorRT components; model, checker, trace, and analysis paths do not. Node names and absolute paths are removed from the review artifact.

Use of generative AI. Generative-AI tools assisted with language editing, organization, bibliography normalization, and consistency-checking scripts. The authors designed the system and model, fixed protocols, executed experiments, inspected failures, and verified every claim against hashed artifacts. Generative AI produced no experimental measurement or ground-truth label.

8 Evaluation

We report the experiments in claim order. Safety and prediction are primary outcomes; throughput and production qualification are independent gates with negative results preserved.

8.1 MPS is a mechanism, not the contract

RQ1 asks whether an active-thread cap can stand in for the radio contract. The answer is no for the tested separate-client and lifecycle conditions.

Cap-Only Diagnostic. A separate-client overrun produced 2/1,500 deadline misses at a 100% cap and 1/1,500 at a 20% cap. Reducing active-thread percentage did not eliminate the tail.

Lifecycle Control. A later matched comparison observed 0/10,000 misses for a CPU socket sham and 12/10,000 for the GPU/MPS-client arm. The result isolates a GPU-client lifecycle effect from the control channel used to trigger it.

Answer to RQ1. These finite samples do not estimate a universal miss probability. They reject the premise that MPS cap selection alone is the certificate. The remaining experiments retain MPS as the execution mechanism and test SoftWall's contract above it.

8.2 End-to-end variable-class qualification

RQ2 asks whether the certificate survives actual NeuralRx, shared conventional recovery, external AI, and radio commit. We evaluate independent development and holdout nodes under one frozen warm mode.

Table 4. End-to-end variable-class qualification.

Metric	Development	Holdout	Combined
Epochs	600	600	1,200
Actual NeuralRx requests	2,400	2,400	4,800
Timely NeuralRx successes	1,750	1,738	3,488
Physical shared recoveries	650	662	1,312
Qwen completions	544	533	1,077
Radio commits	2,400	2,400	4,800
Deadline misses	0	0	0

Qualified Workload. The P180/D155 campaign ran 600 epochs on each node and admitted Qwen contexts from 16 to 512 tokens with class bounds {35, 35, 35, 40, 65, 75} ms. Table 4 reports physical work rather than simulated service.

Whole-Path Results. Maximum observed NeuralRx, recovery, and radio-commit paths were 18.123, 13.465, and 118.022 ms. Every executed Qwen unit remained within its work-class bound. The certificate rejected 55 context-256 and 68 context-512 requests before launch when recovery debt left insufficient room. No Qwen kernel began after its latest-start.

Physical Execution and Answer to RQ2. Separate Nsight Systems campaigns on the same A100, MIG-off, MPS substrate verify physical rather than host-only overlap. A profile of NeuralRx and Qwen contains 191 unique kernel-overlap segments totaling 8.932 ms across seven early-AI windows. A profile of early conventional and other-cell NeuralRx observes overlap in 12/12 windows totaling 12.309 ms. These small profiler-instrumented samples establish kernel co-execution; they do not qualify co-run service bounds or the later end-to-end mode. In that frozen end-to-end campaign, the two nodes executed 4,800 actual TensorRT neural-receiver requests, 1,312 shared-cuPHY recoveries, 1,077 Qwen units, and one commit for every radio request without an observed deadline, bound, guard, credit, or commit violation. This is a finite-sample qualification of the declared warm mode.

8.3 Fault closure

RQ3 asks whether logical recovery remains bound to physical ownership when messages are stale, duplicated, delayed, or lost. We combine finite-state checking with physical fault injection.

State Audit. A finite-state audit checked 51 combinations of epoch, generation, fence, and marker state with no invariant violation. It covers transitions that are difficult to time physically, including reply ambiguity before and after an effect.

Physical Fault Campaign. The A0-A6 campaign covers clean operation, correlated NeuralRx failure, reply delay before and after physical completion, stale and duplicate responses, terminal channel fault, and post-fault continuation. Across qualified nodes it executed 700 epochs, 2,800 actual NeuralRx requests, 942 shared recoveries, and 2,800 radio commits with no observed deadline miss. It included 40 stale/duplicate replay pairs, four terminal channel faults, and 260 continuation epochs after terminal fault.

Fail-Closed Scope. One additional node violated the NeuralRx whole-path bound. We retained it as UQ and did not pool it into the passing result. The evidence supports containment only for the qualified fault and lifecycle classes.

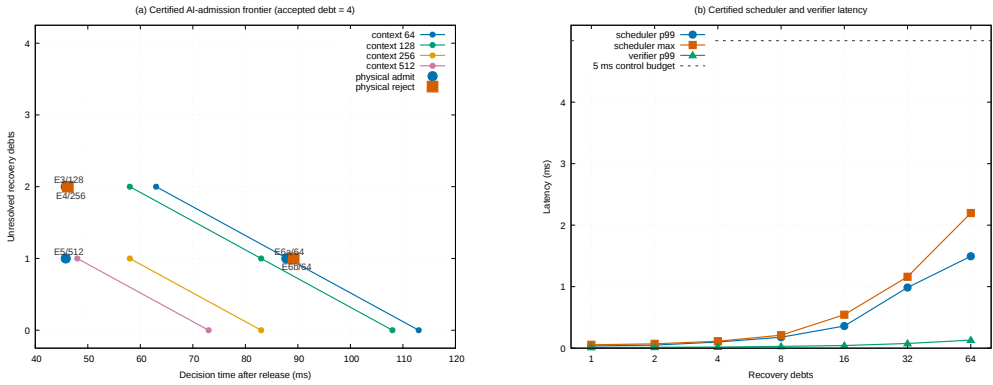


Fig. 2. Qualified AI-admission frontier versus debt, AI class, and decision time (a), and certified-scheduler latency through 64 debts (b). Prespecified physical cases lie on their predicted sides of the frontier. Latencies are finite-run measurements, not WCETs.

8.4 Predicting an unopened physical boundary

RQ4 tests whether the envelope predicts admission before observing a new physical outcome. We compare analytic, exact, retrospective, and prespecified physical decisions.

Model Agreement. The envelope matched exact feasibility on 16,023 qualified states and matched all 1,200 retrospective physical decisions. We then froze seven boundary cases, their order, and node-specific seeds before execution.

Physical Boundary. Across two independent nodes and 180 rounds, model and runtime agreed in all 180. The campaign executed 720 actual NeuralRx requests, 330 recoveries, 90 Qwen units, and 720 commits with no observed deadline miss. Context-64 at the conservative integer decision time of 88 ms was admitted in 30/30 rounds; 89 ms or later was rejected in 30/30.

Answer to RQ4. Figure 2 shows that the same AI class and debt can cross from QSU to QSN as the decision time advances. The envelope predicts this transition without future CRC or realized service time.

8.5 Certified scheduler scalability

RQ5 asks whether certificate construction remains useful beyond the state size handled by exact search. Safety is delegated to the verifier; heuristic incompleteness may reject a feasible state but cannot admit an invalid one.

Exact Comparison. On 600 heterogeneous small states, exact search accepted 504 and the deterministic candidate accepted 494. False-safe was zero and false-conservative was ten. On the current qualified mode, candidate and exact acceptance matched across all 16,023 states.

Control-Path Cost. A 2,800-state grid spans 1-64 debts and capacities 1, 2, 4, and 8. Every returned certificate passed the independent verifier. At 64 debts, candidate decision latency was 1.494 ms p99 and 2.197 ms maximum; verifier p99 was 0.131 ms. These finite-run CPU measurements remain below the evaluated control budget, but they are not WCETs.

8.6 What the certificate prevents

Proposition 2 concerns guarantee authority, not an instruction to manufacture deadline misses. RQ4's prespecified boundary cases provide witnesses using declared service bounds.

Table 5. Safety necessity and AI-first utility are different comparisons.

Policy	All-fail check	AI before debt clears	Finding
Debt-blind current-idle	No	Yes	Cannot guarantee E4/E6b; unsafe diagnostic
Certificate-preserving recovery-first	Yes	No	Safe comparator; 385,262 timely tokens
SoftWall	Yes	Only with witness	Same 385,262 tokens; predicts the boundary

False-Safe Witnesses. In E4, two debts remain at 45 ms and a context-256 request is visible. Mandatory-only recovery is feasible, but debt-blind admission permits a bound-respecting completion at 165 ms, 12 ms beyond the 153 ms guard. In E6b, one debt remains and context-64 becomes unsafe at 89 ms; completion reaches 154 ms, one millisecond beyond the guard. SoftWall classified both states as QSN and rejected all 30 requests per case across two nodes before launch.

Interpretation. These are contract-level counterexamples. We do not label them observed debt-blind deadline misses because the rejected kernels were not launched. The MPS-only results in Section 8.1 provide the separate evidence that physical cap-only tails can occur. Table 5 separates safety authority from AI-first utility.

Bound Sensitivity. The witnesses use the qualified $B_{\text{conv}} = 25$ ms contract. E6b disappears at or below 24 ms; E4 remains until 19 ms, and only E4 remains false-safe for $19 < B_{\text{conv}} \leq 24$ ms. The observed recovery-path maximum was 13.465 ms, or $1.86\times$ below the declared bound, but a sample maximum cannot replace a service contract. A lower bound defines another mode and moves the false-safe interval rather than making unresolved debt irrelevant. That new boundary requires requalification.

A failure-correlation sweep is unnecessary for the safety conclusion because the contract already admits correlated all-fail behavior. Correlation affects expected utility, which RQ6 measures separately.

8.7 Strong baseline and no material headroom

RQ6 tests whether AI-first retiming increases timely external-AI value over a safe policy that already models recovery debt. It does not test whether certificates are necessary.

Matched Comparison. The certificate-preserving recovery-first baseline and SoftWall share max-radio decisions, all-fail admission, observable state, arrivals, service bounds, qualified workers, and single-commit behavior. Their sole semantic difference is whether AI may execute before unresolved recovery physically completes.

Result. Four non-overlapping BurstGPT calibration windows contain 4,290 requests and 1,129,504 offered tokens. Both systems complete 931 timely requests and 385,262 timely tokens. A sensitivity charging every recovery its full 25 ms bound improves SoftWall's timely token value by 0.066%, below the prespecified 5% minimum effect. We did not open the confirmatory throughput holdout.

Answer to RQ6. AI-first retiming does not increase timely tokens on this trace. The evaluated one-second AI SLO allows recovery-first catch up. Conditional capacity and its safety boundary remain observable, but we make no optimizer claim.

Table 6. Lifecycle envelope. “Partial” qualifies only the stated safe subset.

Mode	Status	Claimed property
Warm persistent	Qualified	Full evaluated contract
First request after 30 s idle	Qualified	Bounded first transaction
MPS restart + requalification	Partial	No reuse before requalification
Qwen reload	Partial	Mandatory radio continuity
Same-worker reconnect	Partial	Generation reconciliation
Process/model cold	UQ	None
5 min idle	UQ	None
30 min idle	UQ	None
GC-on	UQ	None
Worker process replacement	UQ	Development canary only

8.8 Lifecycle qualification

RQ7 begins by asking which process and worker lifecycles can use the qualified bounds. The matrix is claim-scoped rather than a checklist to fill after observing results.

Qualified Subsets. The paper has five qualified or partial subsets and five UQ modes. Qualified evidence covers warm persistent execution, the first request after 30 s idle, MPS restart followed by requalification, mandatory radio continuity during Qwen reload, and same-worker reconnect reconciliation. It does not claim optional-service availability during restart or reload.

Unqualified Lifecycles. Process/model cold, 5 min and 30 min idle, GC-on, and worker process replacement remain UQ. A single-node replacement development canary passed its internal gate but lacks an independent holdout. Durable journal recovery would be a new mechanism and remains outside this paper.

8.9 Production exit gates

The synthetic contract does not establish production timing. We use a vendor integration threshold to falsify candidate implementations while keeping target-DU timing as the first unresolved gate.

Timing Contract. Aerial testMAC expects CRC-bearing UL indication by $T_0 + 4.5$ ms. Its validator reconstructs slot T_0 and marks later handler arrival as late. TestMAC is a controlled developer L2 rather than a field-DU d_{MAC} source. The target-DU trace is absent, and we make no production HARQ guarantee.

Path Elimination and Attribution. Under clean, warm, no-AI conditions, local wait-then-recover, same-GPU speculative, and precomputed cross-GPU paths each complete 0/1,000 requests within 4.5 ms. Raw-IQ P2P moves channel estimation through CRC to GPU1 and improves completion to 852/1,000 with 3.831 ms median. Binding cuPHY and caller-owned TensorRT to one stream reaches 885/1,000 but retains 115 late samples. A 300-request profile attributes that path’s tail to remote channel estimation: GPU p50/p99/max are 0.866/4.873/9.592 ms and correlation with pair wall time is 0.9603. TensorRT host enqueue averages 7.443 μ s, while its graph has 0.886 ms GPU p99. Forward/backward P2P copies average 36.261/17.150 μ s. Derate-match host call averages 1,060.105 μ s with 1,105.380 μ s p99, while its GPU p99 is 0.168 ms.

Frozen Fast-Path Gate. We froze persistent Fortran-layout input and stream-ordered assembly as one bounded implementation change. The development gate reaches 995/1,000 timely requests, with 3.813 ms median and 3.995 ms p99; both decoders are correct for all requests. Five late completions reject the candidate and leave the holdout closed. Four are limited by conventional completion

Table 7. Frozen disjoint-seed Sionna CDL-D/E holdout.

Mode	Conv.	NeuralRx	NeuralRx-only	Conv.-only
CDL-D, 100 ns	153/250	164/250	16	5
CDL-E, 100 ns	155/250	163/250	15	7
Paired aggregate	–	–	31	12

and one by remote NeuralRx. Pair latency correlates 0.918 with conventional GPU service and 0.3307 with remote NeuralRx. A conventional profile identifies equalization as its strongest stage correlate ($r = 0.761$; 1.904 ms maximum), while conventional channel estimation reaches 1.295 ms. The remaining tail spans both cuPHY paths. These profiles are mechanism evidence rather than qualification or WCET. The bounded P2 campaign stops at this failed gate. The native C++ path establishes fixture and layout parity only and is not a timing qualification.

8.10 Supported-channel NeuralRx result

The final part of RQ7 asks whether optional NeuralRx provides conditional radio value under a declared external channel contract. Unsupported and supported domains are both reported.

External Aerial TDL-A. Across raw and normalized inputs, public-reference radio settings, one-antenna frequency/time channel execution, and a public-notebook-style FP32 engine, conventional decoding succeeds in every ten-TB arm and NeuralRx succeeds in none. TDL-A remains UQ.

Frozen Sionna Holdout. We reconstruct the Sionna 1.0.2/TensorFlow 2.19 contract. Clean MCS7 and Rayleigh controls pass, and a fixed development matrix selects CDL-D/E at 100 ns while retaining A/B/C failures. A disjoint-seed holdout uses new payload, slot, and channel seeds across five Es/No strata and 50 blocks per stratum per model. Both pipelines pass 50/50 at 10 dB. Table 7 reports all paired outcomes.

Channel Claim. The aggregate discordant outcomes give a two-sided exact $p = 0.00540$. The optional NeuralRx has conditional radio value in this frozen Sionna CDL-D/E mode. The result qualifies neither Aerial TDL-A nor field IQ, and the channel mode has not passed production timing or integrated service-vector qualification.

9 Related Work

Prior systems establish the mechanisms on which SoftWall builds: AI-and-RAN resource allocation, deadline-aware accelerator selection, GPU slack reclamation, and AI/conventional PHY choice. Our comparison asks a narrower question: whether the scheduler represents the unresolved conventional obligation created by an optional receiver for the same TB. Table 8 summarizes this distinction at the level of the protected object.

9.1 AI-and-RAN resource allocation

AI-and-RAN systems already allocate heterogeneous compute across radio and external-AI demand. They establish that dynamic sharing and deadline-aware placement are useful; their public scheduling state does not expose a future same-TB recovery job before that job enters an execution queue.

Shared Acceleration. YinYangRAN adjusts MPS fractions between PHY and ML and can use CPU fallback [10]. CloudRIC places RAN requests across heterogeneous accelerators from queue and deadline estimates [11]. These systems establish MPS-based sharing, fallback, and deadline-aware accelerator selection as prior art. SoftWall does not claim any of those mechanisms.

Table 8. Closest-work distinction. “Not exposed” describes the public design and is not a claim that the mechanism could never be extended.

System	Primary protected/allocated object	State not exposed to its scheduler
YinYangRAN	MPS fraction and CPU fallback	Multi-cell unresolved same-TB recovery calendar
CloudRIC	Queued request and accelerator deadline	Recovery jobs that have not entered a queue
Concordia	Reserved/reclaimed vRAN CPU capacity	Optional-PHY outcome-conditioned debt
DARIS, REEF, XSched, nvtaskset	GPU stage, kernel, context, or partition	Radio recovery, transport ownership, and single commit as one certificate
ARCHES, OCUDU	AI/conventional PHY selection or timing class	Cross-home all-fail debt plus atomic external-AI lease
SMEC	RAN/edge SLO coordination	Same-PHY-GPU optional receiver recovery
SoftWall	Same-TB recovery obligation	Explicitly modeled, certified, and carried through physical completion

Placement and Orchestration. Interplay and CAORA use learning-based orchestration and MIG allocation [12, 13]; HAF separates slow placement from fast deadline-aware CPU/GPU allocation [6]. A distributed AI-RAN platform addresses placement across RAN sites [1]. Their allocation units are workers, resources, or queued jobs. SoftWall represents the conventional recovery of an unresolved optional receiver as mandatory future work tied to one radio result.

9.2 GPU slack and real-time execution

Resource reclamation and GPU execution control explain how lower-priority work can coexist with a latency-sensitive tenant. SoftWall uses these mechanisms, while its certificate protects a radio-specific obligation that spans scheduler state, transport ownership, and physical completion.

Slack Reclamation. Concordia reserves CPU capacity for vRAN and reclaims the remainder [4]. DARIS combines MPS, streams, and stage boundaries to admit lower-priority DNN work [2]. Their reclaimed capacity is derived from a reservation or an execution-stage boundary. Recovery debt changes when an optional outcome resolves and must remain feasible under a correlated all-fail branch.

GPU Execution Control. REEF, XSched, and nvtaskset provide preemption, isolation, or context-scheduling mechanisms [3, 5, 14]. These controls can reduce interference, but they do not by themselves bind an admission decision to launch-time revalidation, endpoint ownership, CUDA completion, and a single radio commit. The seven counterexamples in Section 3 motivate that refinement chain.

9.3 AI/conventional PHY paths and MEC scheduling

Neural/conventional PHY coexistence and 5G-aware application scheduling are also established ideas. The relevant distinction is whether optional decoding creates a multi-cell recovery obligation on the same GPU that also runs external AI.

PHY Path Selection. ARCHES selects neural or conventional PHY experts according to channel conditions [9]. OCUDU exposes in-DU AI-RAN timing classes, retains a conventional path, and validates result and lifecycle behavior [8]. These works rule out novelty claims for neural/conventional

coexistence, fallback, or timing classes. Their public designs do not present the combination of a multi-cell all-fail recovery calendar and an atomic external-AI lease carried to a physical fence.

Edge SLO Coordination. SMEC coordinates application SLOs through decoupled RAN and edge schedulers and uses MPS priority at the MEC GPU [16]. Its AI work executes on a separate edge server rather than the PHY GPU that holds optional same-TB recovery. SoftWall makes no novelty claim for 5G-aware deadline scheduling.

Scoped Difference. Across the public designs reviewed through September 2026, we did not find a system that combines three properties: an executable all-fail calendar for unresolved same-TB recovery, atomic leasing of the remaining capacity to external AI through physical retirement, and a provenance-qualified predictive feasibility envelope. This statement is a scoped literature finding, not proof about unpublished work or work using different terminology.

10 Limitations

The evaluation deliberately separates a qualified synthetic contract from production timing, and a supported simulated channel from unqualified radio domains. These boundaries determine where the current evidence can be used.

10.1 Timing contract and statistical bounds

The measured modes establish finite-sample qualification on one hardware family. They do not convert the synthetic harness into a production radio deadline or a probabilistic sample maximum into a worst-case execution-time bound.

Production Timing. P180/D155 is a synthetic harness contract: P is the inter-release period, not an NR slot duration, and D155 is not a production HARQ deadline. The production d_MAC trace builder, clock and provenance checks, request-specific exact/certified bridge, and expiry validator are implemented and pass 24 unit tests, but a target DU/FAPI trace is unavailable. A new timing vector would require physical requalification. Aerial testMAC supplies a $T0 + 4.5$ ms developer-L2 threshold; the best raw-IQ two-GPU candidate reaches only 995/1,000 clean requests and fails its frozen gate. We make no production HARQ guarantee.

Finite-Sample Qualification. Zero observed violations under a qualified mode do not establish WCET or zero failure probability. Each bound belongs to its hardware, software, placement, workload, and lifecycle fingerprint. A changed fingerprint defines another mode and must pass the same prediction and physical gates.

Hardware Scope. All authoritative hardware results use the NVIDIA A100 family. The four-GPU and multi-home paths exercise NVLink P2P, but the evidence does not establish cross-family or inter-node data-plane validity.

10.2 Fault and lifecycle coverage

The fault model closes bounded response, ownership, and worker-transition cases. Arbitrary GPU or driver hangs and durable recovery after full worker replacement remain outside the qualified set.

Unresolved Long Tail. In C158 attempt 4, one NeuralRx completion reached 350.948 ms in a 400-request run whose NeuralRx p99 was 8.111 ms, a 43.3× excursion; the run consequently had 4/400 deadline misses. Targeted 30 s-idle, quiescent MPS-restart/requalification, Qwen-reload, and same-worker reconnect campaigns did not reproduce the event. Its root cause remains unresolved, and those lifecycle results do not qualify the tail away.

Unqualified Lifecycles. Five lifecycle modes remain UQ, including full worker replacement. The runtime fails closed when a fence or ownership transition is unresolved. Durable journal recovery would introduce a new mechanism rather than complete evidence for the current one.

10.3 PHY and workload generality

The qualified result covers one standardized simulated-channel mode and one representative external-AI trace mapping. It does not establish field-channel generalization or a universal throughput benefit.

Channel Domain. External Aerial TDL-A development screens retain conventional success on every ten-TB arm and NeuralRx success on none across normalization, public-reference radio parameters, exact one-antenna transmission, frequency/time channel execution, and FP32 engine precision. A frozen Sionna CDL-D/E holdout passes its high-SNR pipeline and conditional-value gates over 500 paired blocks, with 31 NeuralRx-only and 12 conventional-only low-SNR outcomes. This qualifies only that simulated CDL-D/E mode; it does not establish Aerial TDL-A, recorded field IQ, or over-the-air behavior. The supported channel has not traversed production timing and integrated service-vector qualification.

External-AI Mapping. BurstGPT supplies real request arrivals and token lengths, while Qwen2.5-1.5B prefill is a representative external workload rather than an AI-RAN domain application. Token bucketing and the one-second deadline are experimental mappings. Different model architectures, generation phases, or independently justified SLOs require new service bounds and replay qualification.

Utility Scope. The strongest safe comparison finds no material throughput advantage. Selecting a tighter AI deadline after observing that result would be post hoc. The demonstrated operational value is advance classification of safe, no-slack, infeasible, and unqualified modes, together with rejection of work that would break a certificate. The paper therefore makes a substrate and envelope claim.

11 Conclusion

Optional NeuralRx creates mandatory future work that current GPU idleness does not expose. SoftWall represents that work as recovery debt, maintains an executable multi-cell all-fail certificate, and admits external AI through a transaction that remains valid across control, transport, launch, and physical completion. Seven measured counterexamples derive the refinement chain, and the predictive envelope classifies a qualified mode before the corresponding physical outcome is opened.

Across two independent A100 nodes, the evaluated modes exercise actual NeuralRx, shared cuPHY recovery, Qwen execution, multi-GPU transfers, and bounded fault transitions. The same evaluation finds no material AI-first throughput gain over a certificate-preserving recovery-first baseline and rejects the production fast-path candidate at 995/1,000. The supported claim is therefore precise: SoftWall provides a conditional-recovery substrate and feasibility envelope for qualified modes; finite samples, unsupported channels, and the absent target-DU timing contract do not establish WCET or a production HARQ guarantee.

References

- [1] Ganesh Ananthanarayanan, Matthew Balkwill, Xenofon Foukas, Zhihua Lai, Bozidar Radunovic, Connor Settle, and Yongguang Zhang. 2025. Distributed AI Platform for the 6G RAN. In *ACM Workshop on Open and AI RAN*. ACM, New York, NY, USA, 15–21. <https://doi.org/10.1145/3737900.3770167>
- [2] Amir Fakhim Babaei and Thidapat Chantem. 2025. DARIS: An Oversubscribed Spatio-Temporal Scheduler for Real-Time DNN Inference on GPUs. In *ACM/IEEE Design Automation Conference*. IEEE, Piscataway, NJ, USA, 1–7. <https://doi.org/10.1109/DAC63849.2025.11132423>
- [3] Joshua Bakita and James H. Anderson. 2025. Hardware Compute Partitioning on NVIDIA GPUs for Composable Systems. In *ECRTS*, Vol. 335. Schloss Dagstuhl–Leibniz-Zentrum fuer Informatik, Wadern, Germany, 21:1–21:25. <https://doi.org/10.4230/LIPICs.ECRTS.2025.21>
- [4] Xenofon Foukas and Bozidar Radunovic. 2021. Concordia: Teaching the 5G vRAN to Share Compute. In *ACM SIGCOMM*. ACM, New York, NY, USA, 580–596. <https://doi.org/10.1145/3452296.3472894>

- [5] Mingcong Han, Hanze Zhang, Rong Chen, and Haibo Chen. 2022. Microsecond-Scale Preemption for Concurrent GPU-Accelerated DNN Inferences. In *USENIX OSDI*. USENIX Association, Boston, MA, USA, 539–558. <https://www.usenix.org/conference/osdi22/presentation/han>
- [6] Haiyuan Li, Yulei Wu, and Dimitra Simeonidou. 2026. Deadline-Driven Hierarchical Agentic Resource Sharing for AI Services and RAN Functions in AI-RAN. arXiv:2605.07547 [cs.NI] <https://arxiv.org/abs/2605.07547>
- [7] NVIDIA Corporation. 2026. *Multi-Process Service Documentation*. NVIDIA. <https://docs.nvidia.com/deploy/mps/>
- [8] Timothy O'Shea, Matthew Pennybacker, and Andriy Kharchenko. 2026. The OCUDU dApp Platform: An Open Runtime and E3 Interface for Real-Time AI-RAN. arXiv:2609.07843 [cs.NI] <https://arxiv.org/abs/2609.07843>
- [9] Neagin Neasamoni Santhi, Davide Villa, Michele Polese, Salvatore D'Oro, Yunseong Lee, Koichiro Furueda, and Tommaso Melodia. 2026. ARCHES: Adaptive Real-Time Switching of AI Models for the RAN. arXiv:2604.23397 [cs.NI] <https://arxiv.org/abs/2604.23397>
- [10] Leonardo Lo Schiavo, Jose A. Ayala-Romero, Andres Garcia-Saavedra, Marco Fiore, and Xavier Costa-Perez. 2024. YinYangRAN: Resource Multiplexing in GPU-Accelerated Virtualized RANs. In *IEEE INFOCOM*. IEEE, Piscataway, NJ, USA, 721–730. <https://doi.org/10.1109/INFOCOM52122.2024.10621380>
- [11] Leonardo Lo Schiavo, Gines Garcia-Aviles, Andres Garcia-Saavedra, Marco Gramaglia, Marco Fiore, Albert Banchs, and Xavier Costa-Perez. 2024. CloudRIC: Open Radio Access Network (O-RAN) Virtualization with Shared Heterogeneous Computing. In *ACM MobiCom*. ACM, New York, NY, USA, 558–572. <https://doi.org/10.1145/3636534.3649381>
- [12] Syed Danial Ali Shah, Maryam Hafeez, Abdelaziz Salama, and Syed Ali Raza Zaidi. 2025. Proactive AI-and-RAN Workload Orchestration in O-RAN Architectures for 6G Networks. arXiv:2507.09124 [cs.NI] <https://arxiv.org/abs/2507.09124>
- [13] Syed Danial Ali Shah, Zeinab Nezami, Maryam Hafeez, and Syed Ali Raza Zaidi. 2025. The Interplay of AI-and-RAN: Dynamic Resource Allocation for Converged 6G Platform. In *IEEE INFOCOM Workshops*. IEEE, Piscataway, NJ, USA, 1–6. <https://doi.org/10.1109/INFOCOMWKSHPS65812.2025.11152939>
- [14] Weihang Shen, Mingcong Han, Jialong Liu, Rong Chen, and Haibo Chen. 2025. XSched: Preemptive Scheduling for Diverse XPU. In *USENIX OSDI*. USENIX Association, Boston, MA, USA, 671–692. <https://www.usenix.org/conference/osdi25/presentation/shen-weihang>
- [15] Yuxin Wang, Yuhan Chen, Zeyu Li, Xueze Kang, Yuchu Fang, Yeju Zhou, Yang Zheng, Zhenheng Tang, Xin He, Rui Guo, Xin Wang, Qiang Wang, Amelie Chi Zhou, and Xiaowen Chu. 2025. BurstGPT: A Real-World Workload Dataset to Optimize LLM Serving Systems. In *ACM SIGKDD*. ACM, New York, NY, USA, 5831–5841. <https://doi.org/10.1145/3711896.3737413>
- [16] Xiao Zhang and Daehyeok Kim. 2026. Enabling SLO-Aware 5G Multi-Access Edge Computing with SMEC. In *USENIX NSDI*. USENIX Association, Boston, MA, USA, 1759–1776. <https://www.usenix.org/conference/nsdi26/presentation/zhang-xiao>